

Three-phase variable reluctance energy harvesting

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ABSTRACT

A variable reluctance energy harvester (VREH) based on electromagnetic induction is developed for generating electrical energy from low-speed rotary motion. The challenge of a VREH at low rotational speeds is not only the low output power, but also the torque ripple that the harvester generates. Cogging torque, the major contribution to this torque ripple, is an inherent characteristic of VREH and is caused by its geometric features. Cogging torque produces acoustic noise and mechanical vibration for a drive system into which the VREH is embedded. This issue is of particular importance at low speeds and with light loads. In this paper, we use an m-shaped VREH as an example to propose a three-phase design in order to reduce the cogging torque but maintain a high output power at low speeds of 5 rpm to 20 rpm. Three identical m-shaped pickup units in a proper arrangement generate high amounts of electrical energy in three phases, but result in a lower torque ripple. Ten prototypes based on the proposed design were fabricated and tested, and their performance were in good agreement with the simulation results. By using the three pickup units in an optimized arrangement, the VREH enhances the energy harvesting performance in comparison to three single pickup units. At the same time, the torque ripple is reduced to one fifth of that produced by a single pickup unit. This demonstrates the strong potential of the three-phase VREH for implementations of self-powered wireless sensing systems in terms of energy output and mechanical effects on the rotary host.

1. Introduction

Energy harvesting is a promising alternative to batteries in low-power electronic systems. A properly designed energy harvester can reduce the time-consuming and cost-intensive maintenance of battery-based systems, especially in remote or inaccessible locations [1–5]. In industrial scenarios, particularly the harvesting of kinetic energy sources has found great attention, with energies originating for example from vibrating and rotating components [6–8]. A challenge has been to harvest significant energies in applications with low rotational speeds. Such rotations can for example be found in feeders, kilns, and drums in industrial applications [9].

Variable reluctance energy harvesting (VREH) has been proposed as a suitable method for harvesting energy at low rotational speeds [10] and several self-powered sensor systems have been proposed based on VREH. Parthasarathy et al. [11] introduced a self-powered wireless wheel speed sensor by reusing an automotive variable reluctance (VR) speed transducer as an energy harvester. Kroener et al. [12,13] investigated the VR principle and found that it is a reliable method for

developing energy harvesting devices for achieving self-powered sensors in railway applications. Moreover, Gong et al. [14,15] designed, optimized and implemented a VREH for a smart bearing system used in rail transport applications. Xu et al. [16] proposed and evaluated an energy-autonomous wireless RPM sensor with an on-rotor VREH in order to implement condition monitoring systems on rotating bodies. Although these studies demonstrated that VREH can be applied in rotating applications, they lacked detailed investigations of the VREH structure and design. Moreover, the studies focused entirely on the output power levels that can be harvested, but neglected how the harvester affects the rotating host.

In [17], Zhang et al. revealed that, during the rotation of a VREH, the generated magnetic torque negatively influences the normal rotational motion of the rotary host of the VREH. More specifically, as presented in [18,19], VREHs induce a strong torque ripple, which can cause undesirable noise and vibrations. For instance, in [19], an optimized VREH with a high output power and power density results in a torque ripple of up to 0.6 N m. Cogging torque is the major component that contributes to the torque ripple, and is a result of the circumferential component of

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magnetic attractive forces between the magnets in the pickup unit and the ferromagnetic toothed wheel [20,21]. This effect is of particular significance at low rotational speeds and with light loads [22]. Thus, the challenge of VREH at low rotational speeds is not limited to low output power levels, but also the torque ripple generated.

In other permanent magnet (PM) machines, the most common technique for reducing the level of cogging torque is the modification of geometrical structures [23,24]. This includes the arrangement optimization of multiple magnets; shape optimization of magnets, slots, and teeth; skewing techniques of magnets, rotor and slots; as well as adding dummy slots and teeth. However, attempts of cogging torque reduction in energy harvesting are limited. Flankl et al. [25] used a shape optimization method to reduce the cogging torque for an electromechanical energy harvester, resulting in a low cogging torque with a peak of 0.1 N m. Na et al. [26] presented a wind energy harvester based on magneto-piezo-elastic effect with an effective arrangement of multiple magnets in the stator and rotor. This enables the energy harvester to operate with a low cogging torque of less than 0.1 N m. Applying these methods successfully enables PM machines and energy harvesting devices to result in low cogging torque. However, cogging torque is an inherent characteristic of VREH and is caused by the slotted structure of the toothed wheel. Applying a skewing technique, such as used in PM machines, on the tooth-slot structure of a VREH would result in a high structural complexity and increases fringing and leakage effects. Moreover, using a shape optimization method may weaken the reluctance effects, resulting in a lower output power. Applying more magnets and keeping a proper distance with the magnets located on the pickup unit may be an effective approach, but is likely to decrease the volumetric power density.

Multi-unit energy harvesting structures have been a popular approach to increase output power and power density in previous research [27–30]. Wang et al. [27] designed an electromagnetic generator consisting of eight stationary coils and a PM array rotated by means of an impact pendulum. The design generated an output power of approximately 225 μ W at a wheel speed of 200 rpm. Ambrozkiewicz et al. [28] connected multiple coils in series resulting in a high voltage and power response of 0.7 V and 10 m W at a pendulum frequency of 15 Hz. In [29], Zhang et al. proposed a rotational energy harvester that converts a linear motion at low frequencies of 0.5 Hz to 5 Hz. Using four magnets results in a higher magnetic coupling and thus in a 4-fold increase in output power as compared to using six magnets. Moreover, a proper connection of multi-phase coils lowers the internal impedance of an energy transducer, thus increasing energy conversion efficiency for the interface circuitry [30]. To the best of our knowledge, no previous study has investigated the effect of multi-unit structures on the torque ripple created in rotational energy harvesting.

In this paper, we propose a three-phase VREH as an example of a multi-unit structure for variable reluctance energy harvesting. The proposed structure uses a pickup unit that was presented in [19], but consists of three units rather than one. The resulting collection of multiple magnets can reduce the cogging torque if the three pickup units are in a proper arrangement. First, we carry out a theoretical study on the proposed structure regarding characteristics of output power, torque ripple, and cogging torque. Based on this theory, an analytical model is developed in order to predict the optimal arrangement of the three pickup units. A three-dimensional finite element analysis (FEA) is performed to compute the performance of the proposed structure in terms of output power and torque ripple. Finally, the simulation results are verified experimentally on a laboratory test bench for a large-diameter shaft of 270 mm and rotational speeds of 5 rpm to 20 rpm. The results demonstrate that the three-phase VREH in its optimized arrangement results in a 3.76-fold improvement in output power and an 80% reduction in torque as compared to a single pickup unit. The three-phase structure with the optimized pick-up unit arrangement thus enables the VREH not only to result in less negative mechanical effects on its rotating host, but also to increase the output power. Consequently, this novel design leads to a significant performance improvement, which

contributes to the autonomous operation of low-energy sensors in rotating environments.

2. m-shaped variable reluctance energy harvester

2.1. Structure

Fig. 1 depicts a single-pickup-unit VREH, which is comprised of one stationary pickup unit and a toothed wheel rotating relatively to the pickup unit. The pickup unit consists of a pickup coil, two identical PMs, and an m-shaped pole-piece. The toothed wheel is a center-hollowed and toothed structure, which is mounted on an existing rotating host such as a rotary shaft. The pole-piece and toothed wheel are made from a soft ferromagnetic material, and both establish a main flux path to reduce the leakage flux and to guide the magnetic flux to pass through the pickup coil. The pickup coil, a rectangular multi-turn solenoid coil, is wound on the center pole of the pole-piece, and its two terminals are coupled to a load or, in most of cases, an energy harvesting interface circuit. An air gap exists between the toothed wheel and the pickup unit, which is a necessary mechanical clearance for the rotation of the toothed wheel. In the pickup unit, the two PMs are placed in close proximity to the air gap at the two outer poles of the pole-piece, facing the air gap with the same pole. This magnet placement creates two flux loops, increasing the rate of flux change in the pickup coil and thus improving the output power of the energy harvester.

2.2. Operation principle

As illustrated by Fig. 1, the rotation of the toothed wheel will lead to a periodic alternation between the two extreme positions, namely the aligned position (Fig. 1), and the unaligned position (Fig. 1b). As a result, the varying reluctance changes the magnetic flux in the pickup coil, and thus it induces an alternating voltage, which can, according to Faraday's law of electromagnetic induction, be expressed as

$$\begin{aligned} V_{\text{open}} &= -\frac{d}{dt} \sum_{n=1}^{N_{\text{coil}}} \phi_i = -\frac{d\theta}{dt} \frac{d}{d\theta} \sum_{n=1}^{N_{\text{coil}}} \phi_i \\ &= -\omega \cdot N_{\text{coil}} \cdot \frac{d\phi}{d\theta}, \end{aligned} \quad (1)$$

where V_{open} is the induced voltage in an open-circuit condition, N_{coil} denotes the number of turns of the pickup coil, ϕ_i is the flux linkage in the i th turn of the pickup coil, θ is the mechanical angular displacement of the toothed wheel, and $-d\theta/dt = \omega$ is the angular velocity. Assuming the flux to be uniformly distributed in the coil, the parameters dominating the induced voltage are the angular velocity, number of coil turns, and magnetic flux change with respect to the angular displacement.

The frequency f_c of the coil voltage is proportional to ω and the number of teeth of the rotating toothed wheel N_{tooth} . Thus, the toothed wheel acts as a mechanical frequency up-converter, and f_c is expressed by

$$f_c = \omega \cdot N_{\text{tooth}}. \quad (2)$$

Applying the maximum power transfer theorem can obtain the power extraction ability of the VREH. Assuming the VREH works at low angular velocities, the inductive component of the pickup coil is neglectable. If the external load resistance R_{Load} is equal to the pickup coil resistance R_{coil} , the load voltage is half of V_{open} , which is expressed as

$$V_{\text{Load}} = \frac{1}{2} V_{\text{open}}. \quad (3)$$

Substituting Eqs. 1 and 3, the maximum attainable power of the pickup coil is

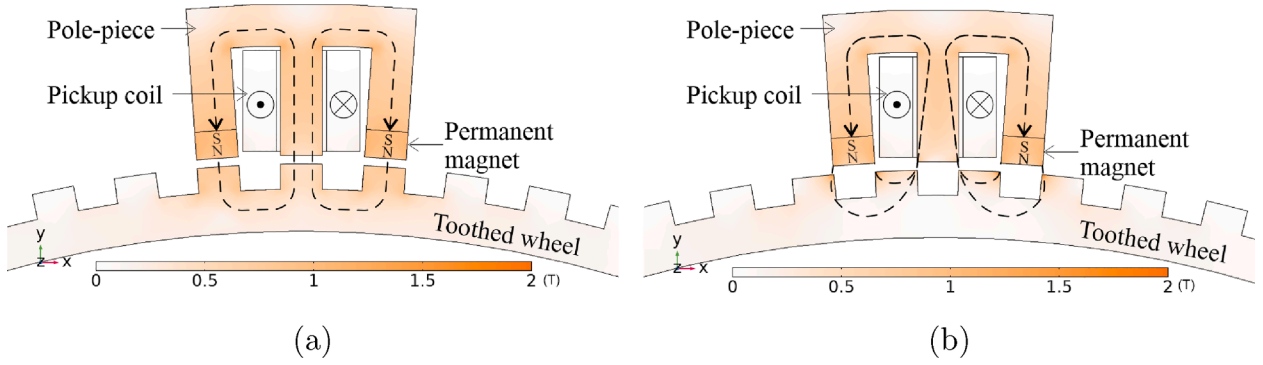


Fig. 1. Structure of the m-shaped variable reluctance energy harvester with magnetic flux density distribution and main flux directions (dashed lines with arrows) in its two extreme positions, (a) aligned position and (b) unaligned position.

$$P_{\max} = \frac{V_{\text{Load}}^2}{R_{\text{Coil}}} \quad (4a)$$

$$= \frac{1}{4R_{\text{Coil}}} \left(\omega \cdot N_{\text{Coil}} \frac{d\phi}{dt} \right)^2. \quad (4b)$$

2.3. Sources of torque ripple

As the pickup unit of a VREH is directly driven by its rotating host, the torque ripple of the VREH is affected by the electromagnetic torque τ_{EM} , the moment of inertia of the rotary host $J \frac{d\omega}{dt}$, as well as the cogging torque τ_{cog} . Thus, by applying the approach of analyzing the torque components of electrical machines [31–33], the torque ripple τ_{ripple} is composed the above three components and is expressed by

$$\tau_{\text{ripple}} = \tau_{\text{cog}} + \tau_{\text{EM}} + J \frac{d\omega}{dt} \quad (5)$$

The electromagnetic torque is directly proportional to the extracted power, and thus a trade-off between power and torque exists. The moment of inertia is mainly defined by the rotating host. This leaves cogging torque, as the main contributor that the harvester design can affect.

As shown in Fig. 1, the pickup unit contains two magnets that point in the same relative direction with respect to the toothed wheel. This results in a cogging torque from each of the magnets, which is in phase with each other. In this paper, we thus use the item τ_{cog} to represent the cogging torque caused by a pickup unit instead of treating each magnet separately. However, in principle, cogging torque in a VREH is produced by the magnetic attraction between the permanent magnets in the pickup unit and the ferromagnetic toothed wheel. It is the circumferential component of the attractive force that attempts to maintain the alignment between the teeth and magnets, which can be expressed by

$$\tau_{\text{cog}} = -\frac{1}{2} \phi_g^2 \frac{d\mathcal{R}}{d\theta}, \quad (6)$$

where ϕ_g is the air gap flux, and $\mathcal{R}$ is the air gap reluctance. As both magnets are in phase in the investigated structure, we can treat them as one single magnet unit for the cogging torque investigation.

Since Eq. 6 has no time-dependency such as angular velocity, the cogging torque is independent of the rotational speed. However, because of $d\mathcal{R}/d\theta$ the air gap reluctance varies periodically during rotation. As a result, the cogging torque is periodic, which can be represented by a Fourier series in amplitude-phase form:

$$\tau_{\text{cog}} = \sum_{k=1}^{\infty} \tau_{N_{\text{slot}} \cdot k} \cdot \sin(N_{\text{slot}} \cdot k \cdot \theta), \quad k \in \mathbb{Z}^+, \quad (7)$$

where N_{slot} is the number of slots, k is the harmonic order, and $\tau_{N_{\text{slot}} \cdot k}$ is the amplitude of the k -th harmonic component.

The electromagnetic torque is generated by the induced current in the pickup coil during the rotation of the toothed wheel. In order to focus on the cogging torque, in this study the electromagnetic torque is minimized by performing simulations and experiments under open-circuit conditions. As a result, there is no current in the coil and thus no electromagnetic torque.

At high rotational speeds, the torque ripple is filtered out by the system inertia. To minimize this effect, in this study, simulations and experiments are conducted under low rotational speed. Together with the open-circuit configuration described above, the observed torque ripple reflects the cogging torque. Thus, Eq. 5 yields

$$\tau_{\text{ripple}} \approx \tau_{\text{cog}} \quad \text{as} \quad \tau_{\text{EM}} = 0 \quad \text{and} \quad \lim_{\omega \rightarrow 0} J \frac{d\omega}{dt} = 0 \quad (8)$$

3. Cogging torque reduction of the three-phase topology

3.1. Theory

The total cogging torque of a VREH with multiple pickup units, $N_{\text{pickup}} \geq 2$, can be computed as the sum of the cogging torque contributed by each pickup unit in Eq. 7, i.e.,

$$\tau_{\text{total}} = \sum_{p=0}^{N_{\text{pickup}}-1} \left\{ \sum_{k=1}^{\infty} \tau_{N_{\text{pickup}} \cdot N_{\text{slot}} \cdot k} \cdot \sin \left[N_{\text{slot}} \cdot k \cdot \left(\theta - p \cdot \beta \right) \right] \right\}, \quad k \in \mathbb{Z}^+, \quad N_{\text{pickup}} \geq 2, \quad (9)$$

where p indicates the p -th pickup unit, β is the angle that each pickup unit is shifted relative to the others, and the trigonometric term is expanded as

$$\sin[N_{\text{slot}} \cdot k \cdot (\theta - p \cdot \beta)] = \sin(N_{\text{slot}} k \theta) \cos(N_{\text{slot}} k p \beta) - \cos(N_{\text{slot}} k \theta) \sin(N_{\text{slot}} k p \beta). \quad (10)$$

Eq. 10 reveals that, depending on the factor β , the relative position of pickup units to each other can modify the cogging torque waveform in a combination of symmetrical and asymmetrical patterns (cosine and sine terms). In other words, the cogging torque may weaken or even cancel one another out by means of the compensation effect, if the multiple pickup units are in an arrangement with an appropriate β . Ideally, an optimized arrangement can achieve a total cogging torque of 0 N m. However, in practice it is not possible to find a value for β that makes Eq. 9 equal to zero for any given value in 0 rad to 2π rad. Alternatively, we can try to eliminate the highest harmonic in Eq. 9, reducing the RMS value of the cogging torque to a minimal value. As a result, only the terms related to $N_{\text{slot}} k p \beta$ remain in the RMS expression of Eq. 9, i.e., [34]

$$\tau_{\text{total-RMS}} = \sqrt{\sum_{k=1}^{\infty} \{ [\mathbb{A}(k)]^2 + [\mathbb{B}(k)]^2 \}}, \quad (11)$$

where

$$\begin{aligned} \mathbb{A}(k) &= \tau_{N_{\text{pickup}}, N_{\text{slot}} k} \sum_{p=0}^{N_{\text{pickup}}-1} \cos(N_{\text{slot}} k p \beta) \\ \mathbb{B}(k) &= \tau_{N_{\text{pickup}}, N_{\text{slot}} k} \sum_{p=0}^{N_{\text{pickup}}-1} \sin(N_{\text{slot}} k p \beta). \end{aligned} \quad (12)$$

Therefore, minimizing the RMS cogging torque in Eq. 11 is equivalent to obtaining an optimal β . A common solution minimizes both terms in Eq. 12. An optimized arrangement for multiple pickup units thus features [35,36]

$$\alpha_{n\pm\beta} = \frac{2\pi \cdot n}{N_{\text{slot}}} \pm \frac{2\pi}{N_{\text{pickup}} \cdot N_{\text{slot}}}, \quad (13)$$

where $\alpha_n = \frac{2\pi \cdot n}{N_{\text{slot}}}$ indicates the n -th normal position (see Section 3.2) for a pickup unit, and $\beta = \frac{2\pi}{N_{\text{pickup}} \cdot N_{\text{slot}}}$ with positive and negative signs representing the two available optimal β with respect to the n -th normal position.

3.2. Optimal arrangement of three pickup units

In this study, we present a procedure for achieving the optimal arrangement, while maintaining a symmetrical distribution of the pickup units. This implementation is divided into two steps. In the first step, we let the three pickup units be in the same relative position with respect to the toothed wheel. All pickup units thus are in a so called *normal position*. An example is shown in Fig. 2a, where all of the pickup units align with some teeth of the toothed wheel. In other words, the deviation angles of the outer pickup units with respect to the middle pickup unit are

$$\begin{aligned} \alpha_{L_n} &= -\alpha_{R_n} = \frac{2\pi \cdot n}{N_{\text{slot}}}, n \in \mathbb{Z} \text{ and } n \geq 3 \\ \alpha_M &= \frac{2\pi \cdot n}{N_{\text{slot}}} = 0, n = 0, \end{aligned} \quad (14)$$

where L, R and M are the left, right and middle pickup units, respectively, $-\alpha_{L_n} = \alpha_{R_n}$ indicates maintaining the symmetrical distribution, the negative sign represents a movement of a pickup unit in clockwise direction, and $2\pi \cdot n / N_{\text{slot}}$ is their corresponding n -th normal positions they shift to.

The angular distance between two adjacent normal positions is defined as *pitch* in this paper, which is expressed as

$$\text{pitch} = \frac{2\pi}{N_{\text{slot}}} = \frac{360\text{deg}}{90} = 4 \text{ deg}. \quad (15)$$

Since every pickup unit occupies 2.5 pitches, as illustrated in Fig. 2a, the first available normal position for both left and right pickup units starts from $n = 3$.

In the second step, as shown in Fig. 2b and 2c, we only move the left and right pickup units with respect to the middle one but maintain their symmetrical distribution, resulting in the same shift angle but in opposite directions. According to Eqs. 13,14, if $\alpha_M = 0$, the left and right pickup units have two pairs of deviation angles with respect to their normal position that minimize the cogging torque. For the first pair, both pickup units are shifted towards the middle pickup unit (see Fig. 2b):

$$\begin{aligned} \alpha_{L3-\beta} &= \frac{2\pi \cdot 3}{N_{\text{slot}}} - \beta = \frac{8\pi}{135} \approx 10.67\text{deg}, \\ \alpha_{R3+\beta} &= \frac{2\pi \cdot 3}{N_{\text{slot}}} + \beta = \frac{8\pi}{135} \approx -10.67\text{deg}, \\ \alpha_M &= 0\text{deg}; \end{aligned} \quad (16)$$

for the other one they are shifted away from it (see Fig. 2c):

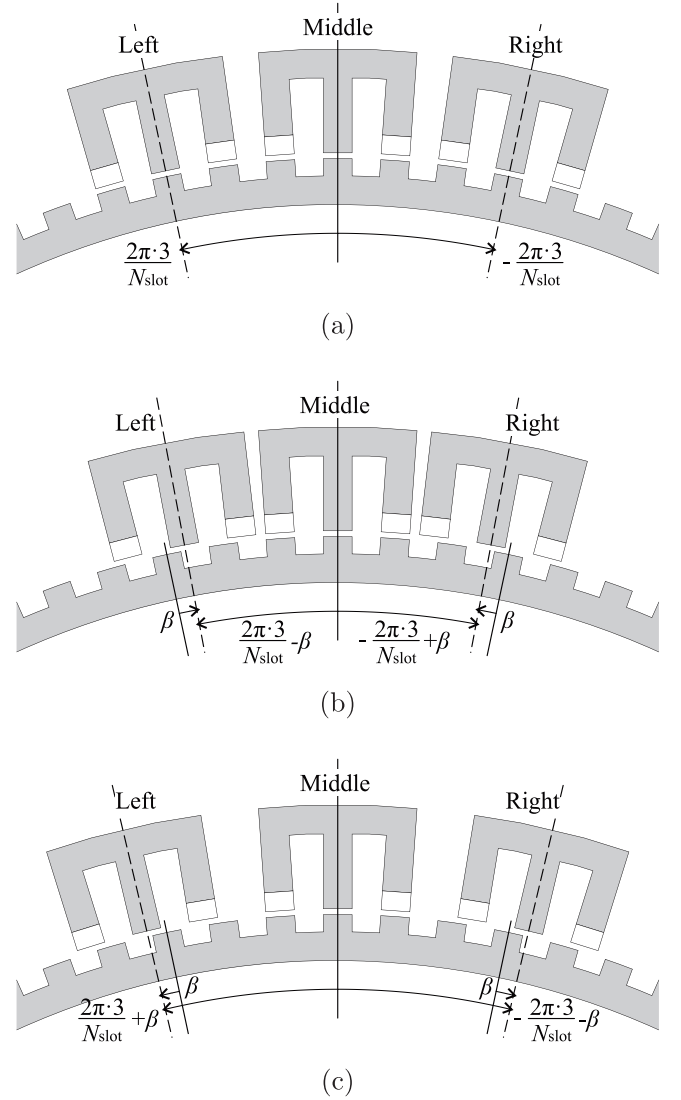


Fig. 2. Three pickup unit arrangements in (a) normal position (b) optimal phase-shift with small distance between pickup units and (c) larger distance between pickup units.

$$\begin{aligned} \alpha_{L3+\beta} &= \frac{2\pi \cdot 3}{N_{\text{slot}}} + \beta = \frac{2\pi}{27} \approx 13.33\text{deg}, \\ \alpha_{R3-\beta} &= \frac{2\pi \cdot 3}{N_{\text{slot}}} - \beta = -\frac{2\pi}{27} \approx -13.33\text{deg}, \\ \alpha_M &= 0\text{deg}. \end{aligned} \quad (17)$$

Moreover, Fig. 3 shows the theoretical cogging torque as a result of different pickup unit arrangements. With varying pitches, the three pickup units can lead to a different cogging torque. The two optimal positions as previously described can be clearly identified and repeat for every tooth. In addition, if the three pickup units are in normal positions, the cogging torque from all pickup units are in phase. As a result, this arrangement produces the highest total cogging torque. If the left and right pickup units shift to the half-way point between two adjacent normal positions, i.e.,

$$\alpha_{L(2n+1)} = -\alpha_{R(2n+1)} = \left[\frac{2\pi \cdot n}{N_{\text{slot}}} + \frac{2\pi \cdot (n+1)}{N_{\text{slot}}} \right] / 2 = \frac{\pi \cdot (2n+1)}{N_{\text{slot}}}, n \in \mathbb{Z} \text{ and } n \geq 3, \quad (18)$$

the VREH produces a second peak in cogging torque.

The results in Fig. 3 are based on a simplified calculation for Eq. 12:

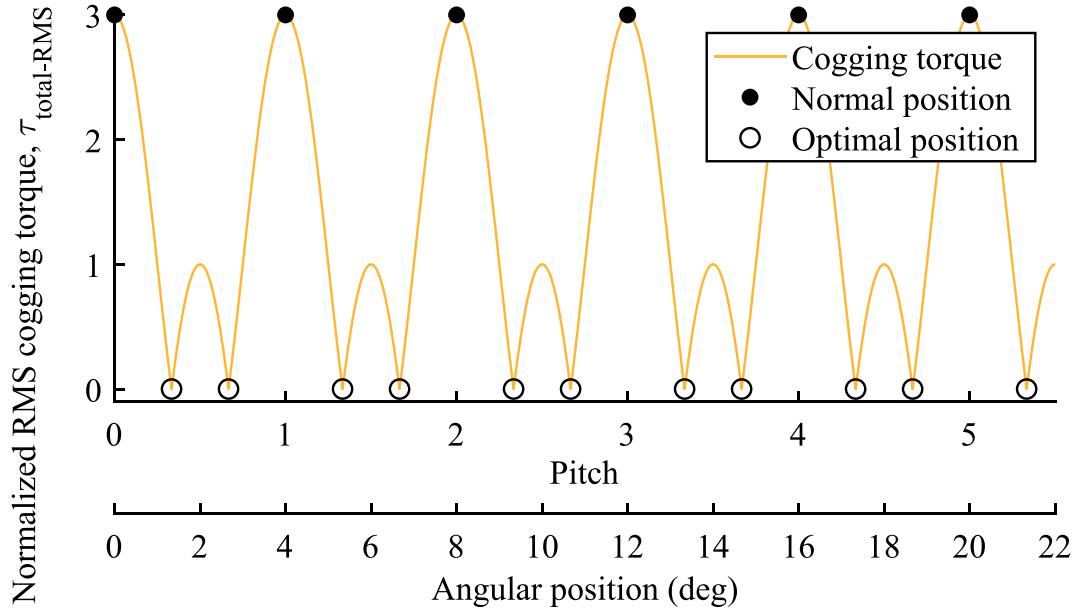


Fig. 3. Analytical estimation of the normalized cogging torque versus angular shift angle between the three VREH pickup units.

(a) we assume there is no additional effect on the cogging torque and thus use the first harmonic, i.e., $k = 1$; (b) we normalize the amplitude, i.e., $\tau_{N_{pickup}, N_{slot}, k} = 1$. Thus, Fig. 3 can provide us with the optimal arrangement, but can not estimate the exact cogging torque amplitude. The exact amount of cogging torque can be determined by FEA simulation and experiment as presented in following sections.

4. Finite Element Analysis setup

Finite element analysis (FEA) can accurately predict cogging torque taking into account magnetic saturation and complex structures [24]. In this study, a 3D time-dependent FEA of the proposed design is carried out using the COMSOL Multiphysics AC/DC module, and the corresponding numerical and graphical results are obtained. Using the time-dependent study and 3D modeling in COMSOL the non-linearity of magnetic properties (B-H curves) of materials and the end effects of the VREH are taken into account.

The torque ripple of each VREH structure is estimated according to Arkkio's method [37], which is expressed as

$$\tau = \frac{2}{\mu_0 \cdot l_{AG}} \cdot \int_0^{l_{free}} \int_S B_r \cdot B_\phi \cdot r_{AG} dS dz \quad (19)$$

Herein, l_{free} is the depth of the simulated free space domain, l_{AG} is the air gap length, S is the surface along the air gap (in the xy-plane), r_{AG} is the air gap radius (i.e., the distance from the center of the shaft to the center of the air gap), and B_r, B_ϕ are the radial and tangential components of the flux density in S , respectively. The factor of two compensates for the fact that only half of the symmetrical model in z-axis direction is simulated as shown in Fig. 4a.

In the FEA, the torque calculation is sensitive to the meshing of the air gap, especially in the vicinity of the slot opening, which is the region that requires fine discretization as shown in Fig. 4b. As a result, the completed meshing consists of 1155018 domain elements, 192469 boundary elements, and 8531 edge elements.

5. Experimental setup

In this section, the prototypes and setup for the experimental verification of the proposed models are described. We use a power-optimized structure of the m-shaped VREH as shown in Fig. 5 [19] for the experimental study.

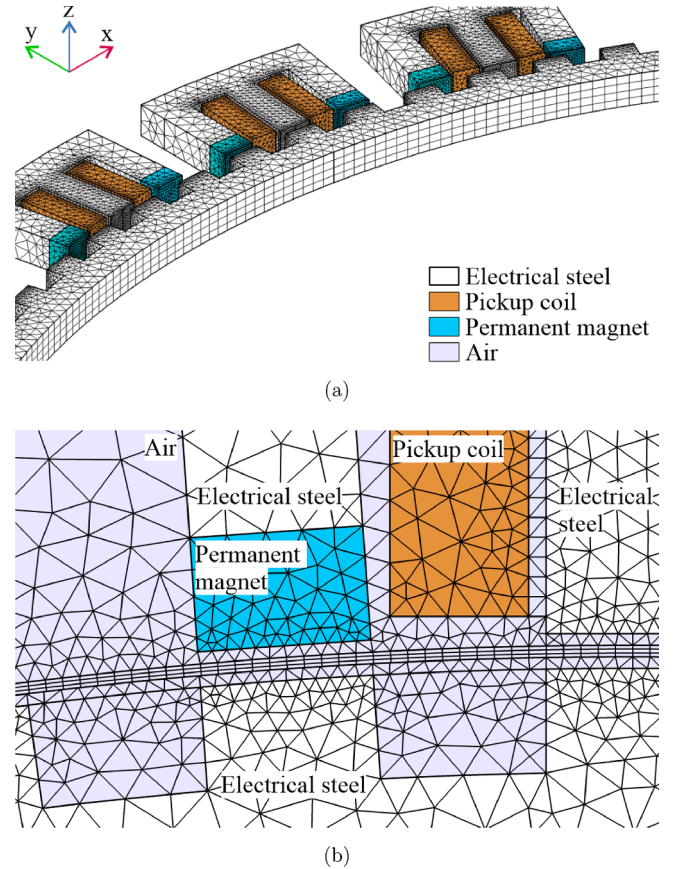


Fig. 4. Mesh configuration of the VREH in the FEA (a) without the air domain and (b) in a close-up view of the air gap.

5.1. Pickup unit prototypes

As exemplified in Fig. 6, the three identical pickup units are clamped in acrylic housings with different arrangements. Each of the pickup units consists of a m-shaped pole-piece and two opposing bulk magnets. The

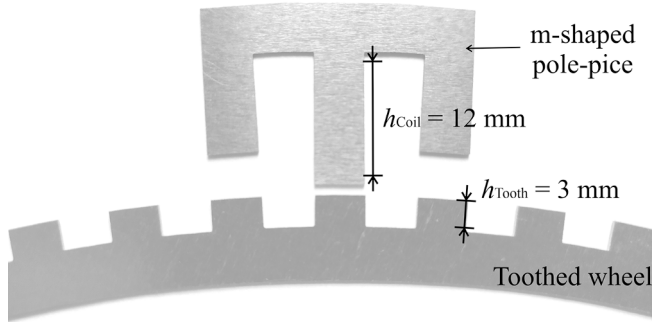


Fig. 5. Single layer of the laminated, m-shaped pole-piece and toothed wheel, where h_{Coil} is the pickup coil height, and h_{Tooth} is the tooth height (or slot depth) of the toothed wheel.

pole-piece is comprised of a 28-layer lamination of electrical steel (NSC-35H230), and N50-NdFeB magnets are used. Ten different prototypes with different arrangements (itches) are produced, and the angular shift between pickup units for each prototype is listed in [Table 1](#).

5.2. Rotating environment and measurement setup

The experimental setup, as illustrated in [Fig. 7a](#), consists of four subsystems, namely a drive unit, a torque measurement setup, a large-diameter shaft, and a support for the VREH prototypes. The details of all units are listed in [Table 2](#). The drive unit spins the shaft and the attached torque sensor and rotary encoder, implementing a drive-measurement-load system. The drive unit, consisting of a gearbox and a DC motor, is operated by a computer using a motor controller unit. The measurements of torque and rotational speed are sampled by a data acquisition (DAQ) device. The DAQ digitizes the measurements and forwards them to the computer. [Fig. 7b](#) shows a close-up view of the air

gap, the toothed wheel and the three pickup units. The pickup units are supported by a multi-axis linear lift stage. Adjusting the linear stage ensures that each pickup unit has an identical air gap to the toothed wheel. The air gap is accurately set to 1mm and verified by a feeler gauge.

The pickup coil poses a large inductive component, and thus ideally a complex conjugate impedance match between the coil and load is necessary to achieve maximum power transfer. However, the VREH operates in this study at low rotational speeds (i.e. 5rpm to 20rpm), which results in low electrical frequencies. Therefore, as shown in [Fig. 8](#), the load circuit is simplified to be purely resistive.

5.3. Cogging torque measurement

Cogging torque is especially prominent at lower speeds, because the shaft's moment of inertia under these speeds is low. Therefore, the torque measurement in this study is carried out at a rotational speed of 0.36rpm, which is the lowest speed that the drive unit is able to operate at. Moreover, according to [Eq. 6](#), different rotational speeds only change

Table 1
Specification of the ten prototype arrangements.

No.	Pitch	Angular shift (°)	Notes
1	2.625	10.50	
2	2.675	10.70	close to an optimal position Eq. 16
3	3.000	12.00	in normal positions
4	3.250	13.00	
5	3.333	13.33	close to an optimal position Eq. 17
6	3.375	13.50	
7	3.675	14.70	close to an optimal position
8	3.850	15.40	
9	4.000	16.00	in normal positions
10	4.250	17.00	

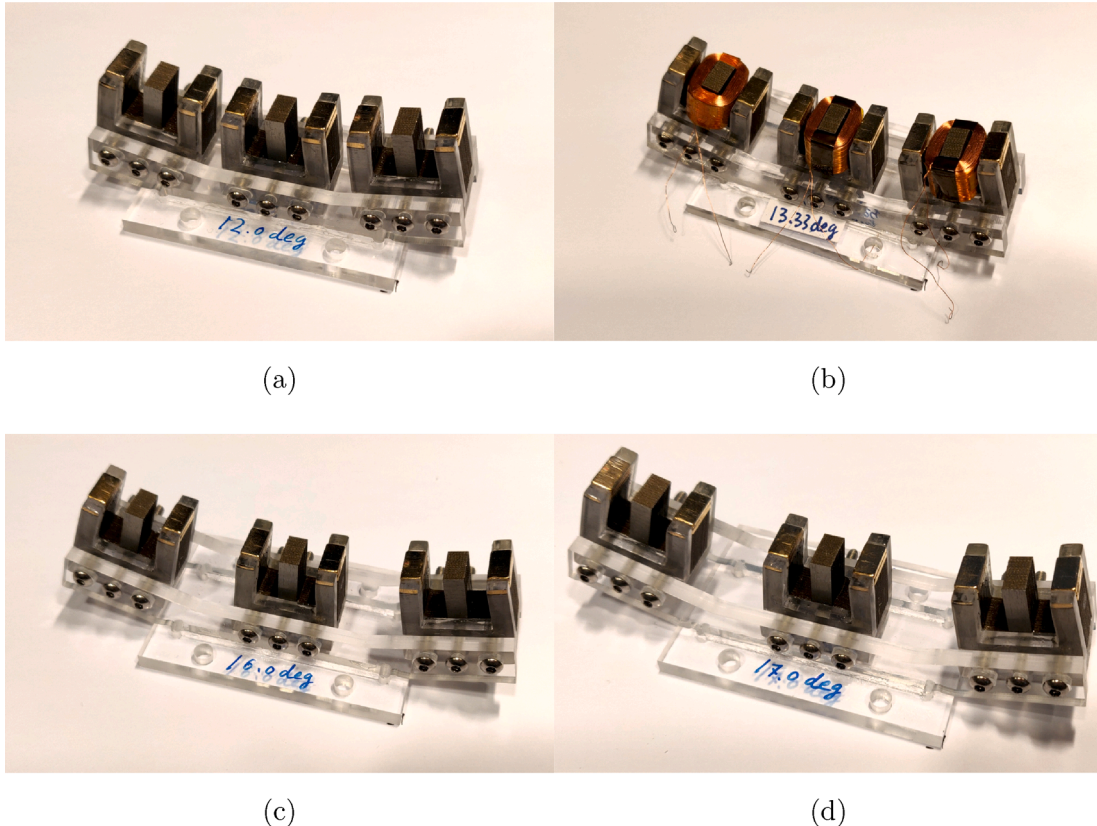


Fig. 6. Prototypes of the three-phase VREH with different pickup unit arrangements: (a) 12.0deg, (b) 13.33deg with pickup coils, (c) 16.00deg, and (d) 17.00deg.

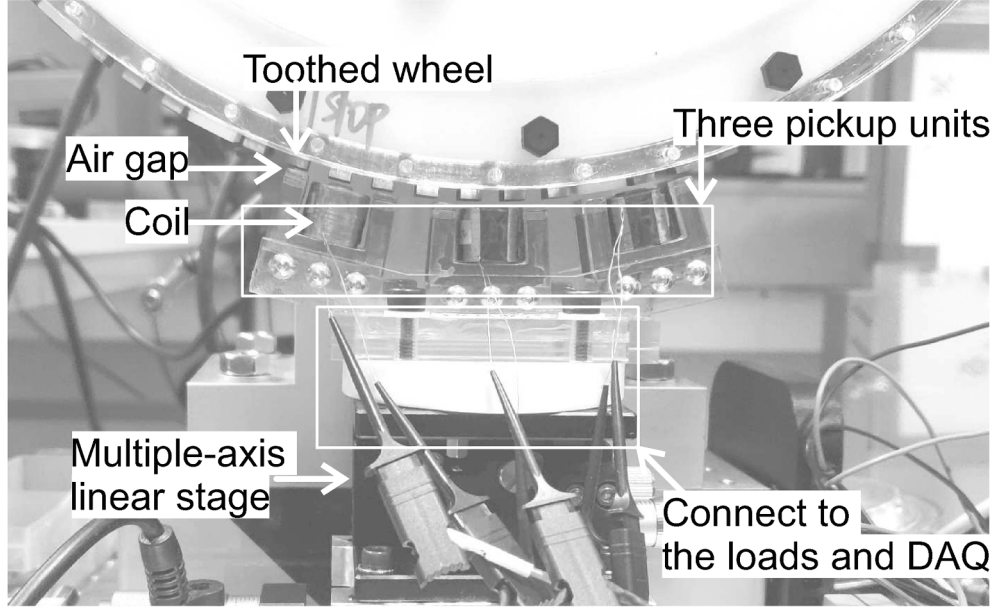
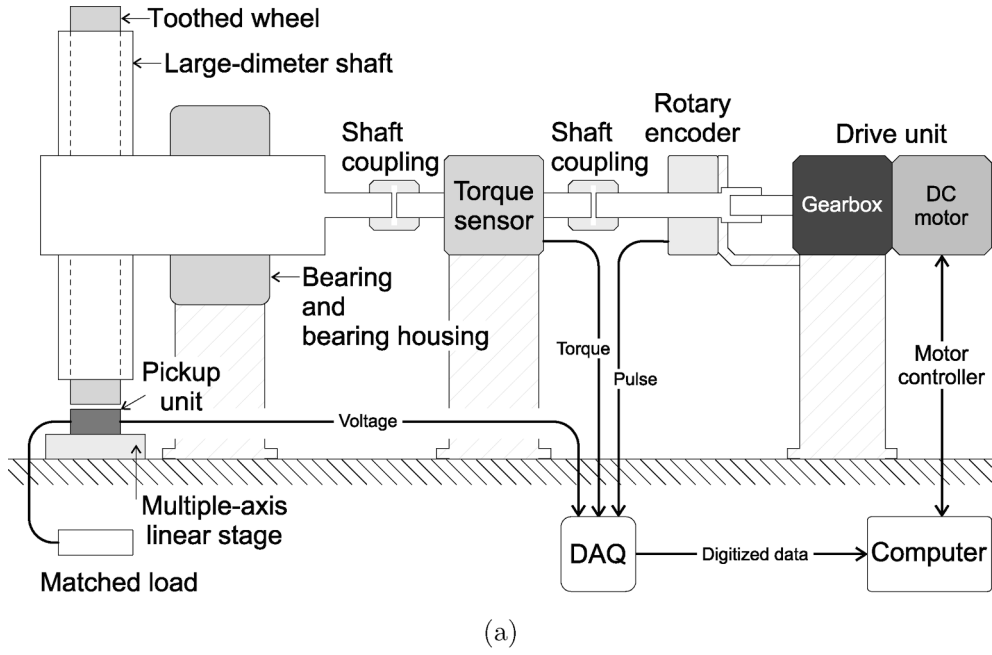


Fig. 7. Experimental setup in (a) a schematic overview and (b) a close-up view of the pickup unit.

the oscillation period of the cogging torque, but do not affect its amplitude and waveform profiles. In other words, the torque is only dependent on the relative mechanical position of the pickup unit and toothed wheel. Thus, the experimental measurements obtained at a low speed even provide the cogging torque waveform and peak values for higher rotational speeds.

6. Results and discussion

In this section, we present and compare results of simulations and experiments to verify the analytical estimation of the optimal arrangement, and to obtain values for torque and output power. First, we compare the numerical simulation with the experimental results to verify that the FEA is a reliable method. Second, we then use the verified

FEA setup to further investigate the three-phase VREH performance and confirm the validity of the analytical model on the optimal arrangement prediction. Finally, the optimal arrangement of the three-phase VREH is compared to a single-phase VREH with respect to output power and torque ripple.

6.1. Experimental verification of the FEA

Fig. 9 exemplifies the time-dependent output voltage of each pickup coil in the three-phase VREH obtained by simulation and experiment. The output voltage can be used to calculate the instantaneous power according to

$$P(t) = \frac{V(t)^2}{R} \quad (20)$$

Table 2
Specification of the experimental setup.

Unit	Key parameters	Notes
DC motor	Rated speed 3000rpm	Brushless DC motor 48V, 500W
Gearbox 1	Planetary gearhead Reduction 30:1	For spinning at 5–20 rpm (power measurement)
Gearbox 2	Planetary gearhead Reduction 216:1	For spinning at 0.36rpm (Torque measurement)
Torque sensor	Range: $\pm 5\text{N m}$	Linearity deviation 0.5% in the full scale
Rotary encoder	Resolution: 2048 ppr	ppr: pulses per revolution
DAQ	12-bit ADC	Saleae Logic Pro 16
Large-diameter shaft	Diameter: 270mm	
Magnet	NdFeB-N50	Dimension (mm) $3.5 \times 5 \times 10$
Coil	Inductance: 210mH DC resistance 72 Ω	Copper wire Number of turns: 1567
Pole-piece	NSC-35H230	m-shaped design[19]
Toothed wheel	Electrical steel sheets NSC-35H230	Number of teeth: 90

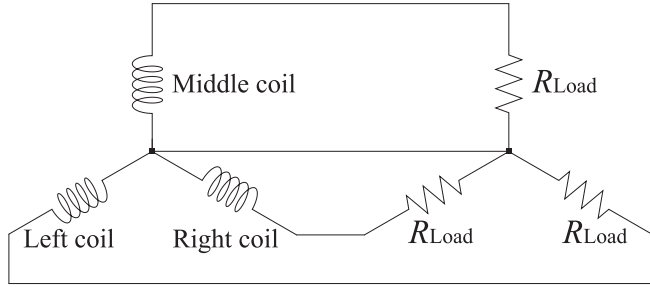


Fig. 8. Circuit implementation of the three coils and their load in COMSOL and experiment.

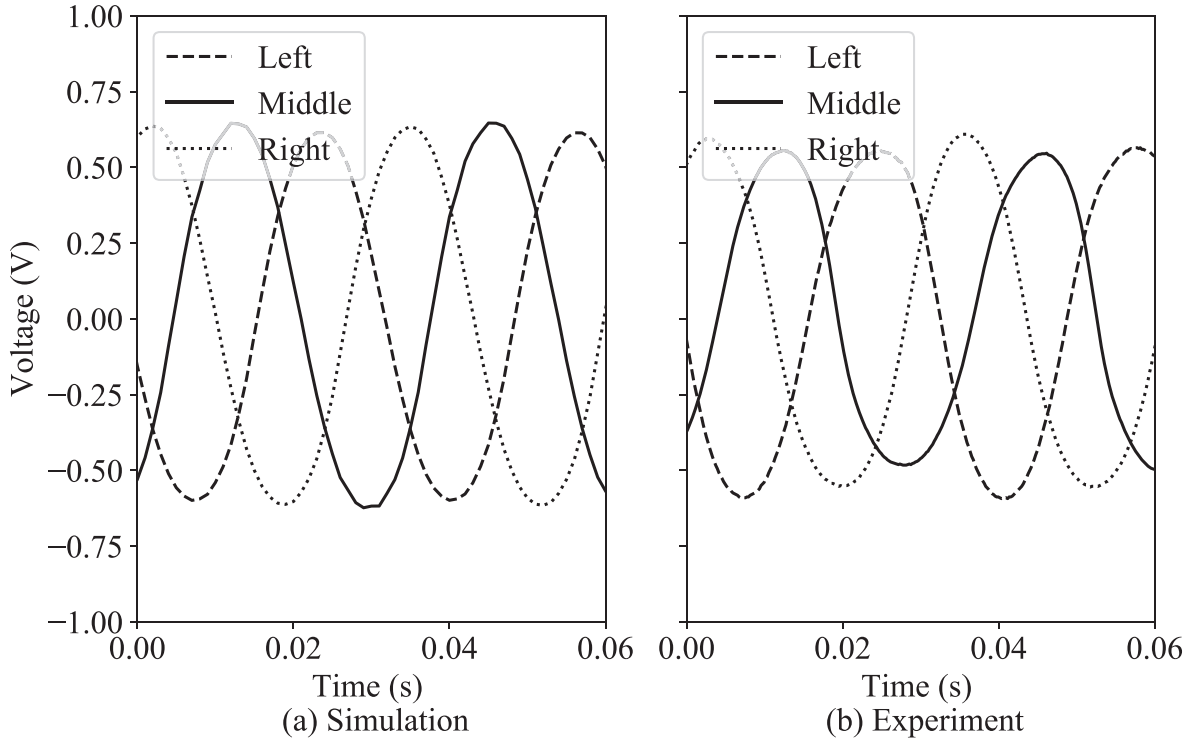


Fig. 9. Voltage output of the arrangement with 13.33deg angular shift under the maximum power transfer condition and a rotational speed of 20rpm: (a) Simulation and (b) experiment.

where $V(t)$ and $P(t)$ are the voltage and power with respect to time t , respectively, and R is the load resistance. Thus, the RMS output power in both the simulation and experiment is obtained by

$$P_{\text{RMS}} = \frac{1}{T} \frac{\Delta t}{2N} \sum_{n=0}^{N-1} [P(t_n) + P(t_{n+1})], \quad (21)$$

where T is the signal period, Δt is the step size, N is the number of samples, and $P(t_n)$ is the instantaneous power with respect to the time shown in Eq. 20. However, in the experiment, the RMS output power in each period of oscillation (i.e., $1/f_c$) varies due to small air gap variations in the mechanical setup. In order to minimize the influence of these variations, the experimental measurements of P_{RMS} are conducted over 50 mechanical rotations of the toothed wheel, resulting in an average output power over 4500 electrical cycles ($= 50 \text{ rotations} \times 90 \text{ teeth}$). Based on the above calculation, Table 3 lists the output power of each prototype and the respective equivalent from simulation. The results demonstrate a good agreement between the simulated and measured output power, with error values of below 7% and a mean absolute percentage error (MAPE) of approximately 4%.

Fig. 10 depicts the instantaneous torque ripple that different arrangements result in. Together with the results listed in Table 4, the waveform and peak of the torque ripple are influenced significantly by the pickup unit arrangements. Moreover, Table 4 shows the relative error between the simulation and experiment for the torque ripple estimation, with a MAPE of approximately 10%. The relatively large errors are a consequence of the relative small torque amplitudes. Moreover, it can be observed that the FEA tends to overestimate the torque.

When the three pickup units are in phase (i.e. at Pitch 3.000 and 4.000), the peak torque ripple and relative error is the largest. At these torque values a significant vibration is induced in the experimental setup, including the pickup units fixed on their support and toothed wheel. As a result, the mechanical vibration causes the air gap between the pickup units and toothed wheel to fluctuate and to differ from its

Table 3
Power output comparison between simulation and experiment.

Pitch	Middle coil output power (uW)		
	FEA	Experiment	Error (%)
2.625	639.734	601.379	6.38
2.675	651.174	611.128	6.55
3.000	788.257	767.495	2.71
3.250	726.649	696.917	4.27
3.333	686.683	671.099	2.32
3.375	669.534	655.263	2.18
3.675	640.929	602.378	6.40
3.850	683.511	643.527	6.21
4.000	702.653	676.506	3.87
4.250	655.892	633.600	3.52
MAPE			4.44

ideal of 1mm. This results in a deformed waveform of the torque ripple in the experiment as compared with the ideal sinusoidal waveforms obtained from the FEA (cf. Fig. 10a and 10e). In contrast, for the other arrangements in Fig. 10, the cogging torque is significantly weakened, which also reduces the error between simulation and experiment. Overall, however, simulation and experiment show the same trend and the FEA provides a sufficiently high accuracy in estimating the torque ripple of the different arrangements. From these results it can be observed that the optimal arrangement is that of pitch 3.333, producing the lowest torque ripple of 0.114N m and 0.115N m obtained in simulation and experiment, respectively.

6.2. Performance of the three-phase VREH

Applying multiple pickup units to a VREH does not change the reluctance variation ($-d\mathcal{R}/d\theta$ in Eq. 6) of each pickup unit. However, multiple pickup units introduce more magnets, leading to a larger flux change in the coils if these magnets are in a proper arrangement [18]. Thus, according to Eq. 4b, the output power increases quadratically with the flux change if the VREH operates at a constant rotational speed ω . Fig. 11 depicts the RMS power generated in each coil for different arrangements of the three-phase VREH. From the figure it is obvious that the middle coil produces a higher output power than the outer coils. This can be explained by the fact that the magnets in each pickup unit also increase the flux density in the air gap of pickup units adjacent to it. In other words, a portion of the leakage flux of one pickup unit contributes to the flux of the neighboring pickup units, which is beneficial for power generation. As the middle coil has two neighboring pickup units, this effect is stronger for it than for the outer pickup units, which only have one direct neighbor.

It can also be observed from Fig. 11 that this aid between pickup units is dependent on the pickup unit arrangement. The effect is strongest when pickup units are in phase (i.e. pitch 3.0 and 4.0 in the figure), and weakest half-way between these arrangements (e.g. pitch 3.5). In some arrangements (e.g. pitch 2.625, 3.575, and 4.5), the effect can be negative and thus reduces the output power of the neighboring pickup unit. This is for example the case when the leakage flux of the outer pickup units mainly contributes to the flux of the middle unit in its unaligned position, and thus reduces flux change between aligned and unaligned positions. Moreover, the influence between pickup units is weakened by a low air-gap flux density caused by an increased distance between two adjacent pickup units. This is demonstrated by the downward trend of output power with increasing pitch in Fig. 11.

Fig. 11 also shows the peak torque produced by each arrangement. The resulting waveform is in good agreement with the analytical results shown in Fig. 3. The minimal torque ripple is produced by the arrangements of 10.67deg, 13.33deg, as well as its equivalents at higher distances between the pickup units. Comparing output power and torque, a trade-off situation can be observed, because maximum output power and minimum torque are obtained with different arrangements.

Based on the ratio of power and torque from the experimental measurements shown in Table 5, a pitch of 3.333 can be identified as an optimal solution.

As a high output power is of significant importance in energy harvesting applications, both power and torque should be taken into account when determining a proper arrangement of the pickup units. In Fig. 11, the range of 12.00deg to 13.33deg is the most suitable, defining the boundaries for highest output power and lowest torque. Fig. 12 visualizes the trade-off between output power and torque in this range. With a lower pitch, a higher output power can be achieved, while a higher pitch reduces the torque. Consequently, this figure can act as a guide for selecting an acceptable torque ripple and predict the resulting output power, or vice versa.

6.3. Comparison to a single-phase VREH

Fig. 13 depicts the RMS output power of every coil, as well as the total power of the combined three-phase pickup unit, for the arrangement resulting in lowest torque (i.e. pitch 3.333, 13.33deg) as a function of angular velocity. The estimated output power can provide sufficient power levels for duty-cycled, low-energy wireless sensor systems. For example, the total output power reaches 1.92m W at a rotational speed of 10rpm, which is a 3.76-fold improvement when compared to the 0.51m W output power of a single pickup unit under the same excitation [19]. For a wireless sensor system such as that proposed in [16] this increases the sample rate from 1Hz to 11Hz, assuming a relatively low power extraction efficiency (PEE) of 60%. Moreover, the selected pickup unit arrangement enables the VREH to significantly reduce the cogging torque, resulting in a torque ripple with a peak of 0.11N m, which is less than 20% of torque generated by the single-phase VREH (see Fig. 14).

6.4. Comparison to a single-phase VREH with magnetic compensation

An alternative approach for the reduction of the cogging torque is to compensate for the cogging torque of the pickup unit by additional magnets. Using the same shifting method as described for the three-phase design, the optimal position for the additional magnets can be found. The structure of this design after optimization is shown in Fig. 15.

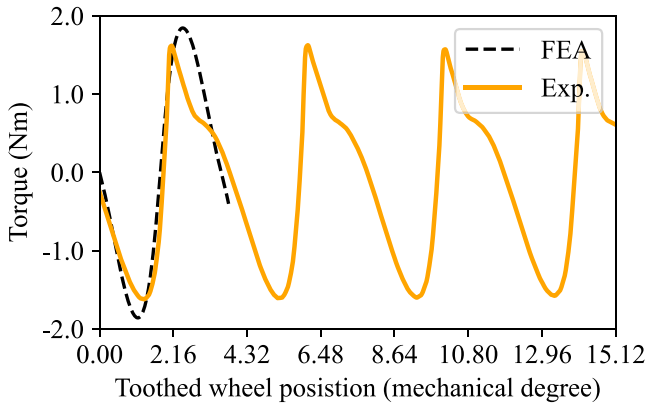
To obtain the optimal structure two parametric sweeps have been conducted in COMSOL: (i) a sweep of angular distance, and (ii) a sweep of magnet height. In other words, we first identify the optimal position of the two magnets (see Fig. 16), and then determine their optimal height (see Fig. 17).

Based on this investigation, a maximum power-to-torque ratio of 4045 μ W/N m at a magnet height of 24mm and magnet positions at a pitch of 2.315 was found. In comparison, this is a power-to-torque ratio of less than a fourth of the 16705 μ W/N m obtained with the optimized three-phase design. This demonstrates the superiority of the proposed solution.

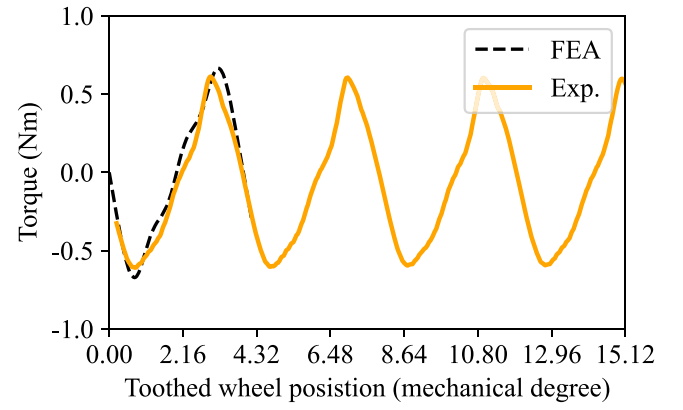
7. Conclusions

In this paper, we proposed, implemented and evaluated a VREH with multiple pickup units for rotational kinetic energy harvesting. The multi-unit structure consists of three identical m-shaped pickup units, resulting in a VREH with three-phase electrical power output. An analytical method based on magnet-shifting has been proposed to determine the optimal arrangements of the three pickup units and was verified by numerical simulations and experiments. Moreover, the obtained results were compared to those of a VREH with single pickup unit, with and without additional, torque-compensating magnets.

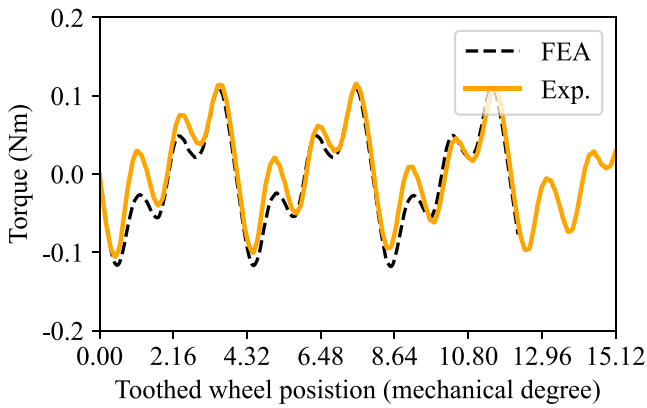
The results show that the optimized three-phase VREH leads to a 3.76-fold power gain in comparison to the single-phase VREH, while generating less than 20% in peak torque. The power-to-torque ratio is with 16705 μ W/N m more than 4 times higher than that of a single VREH with additional, torque-compensating magnets. Numerical simulations



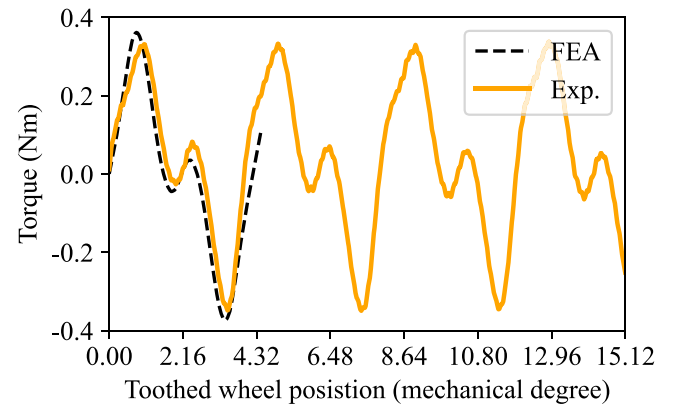
(a) Pitch 3.000 (12.00 deg)



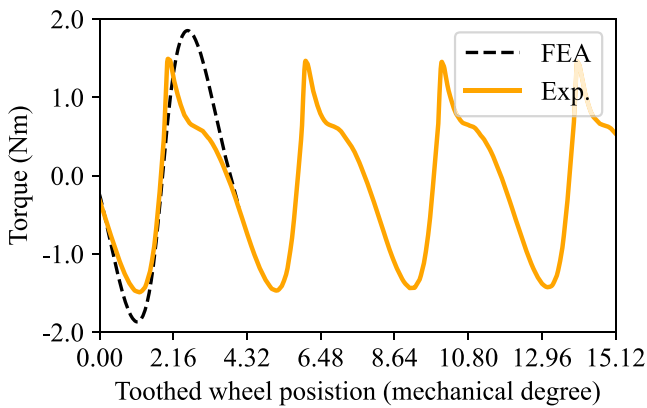
(b) Pitch 3.250 (13.00 deg)



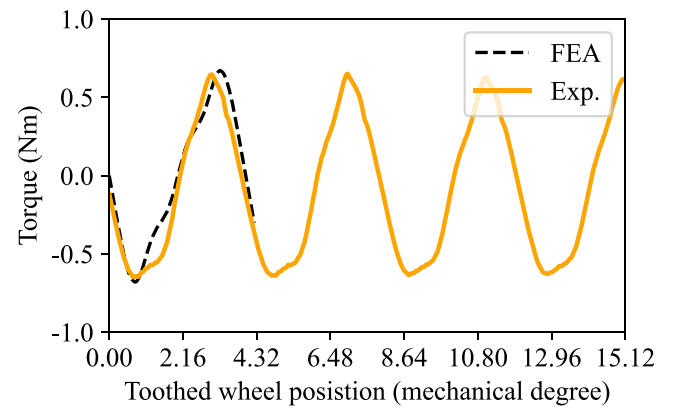
(c) Pitch 3.333 (13.33 deg), optimal structure



(d) Pitch 3.375 (13.50 deg)



(e) Pitch 4.000 (16.00 deg)



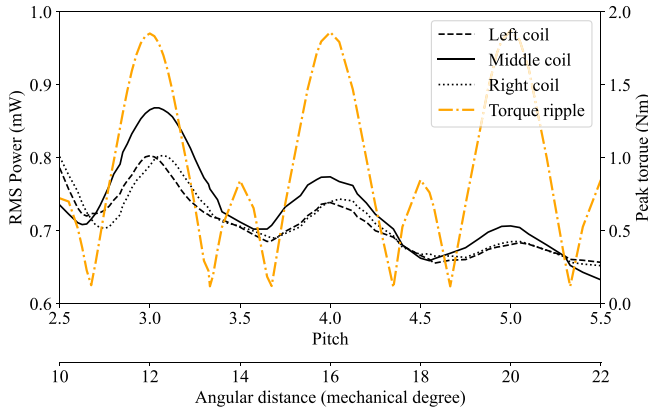
(f) Pitch 4.250 (17.00 deg)

Fig. 10. Comparison of experimental and simulation results of instantaneous torque ripple as a result of different arrangements (a)-(f) at a rotational speed of 0.36rpm.

Table 4

Torque ripple comparison between simulation and experiment.

Pitch	Peak torque ripple (Nm)		Error (%)
	FEA	Experiment	
2.625	0.3841	0.3637	5.61
2.675	0.1265	0.1101	14.90
3.000	1.8515	1.6224	14.12
3.250	0.6686	0.6103	9.55
3.333	0.1141	0.1152	0.95
3.375	0.3677	0.3408	7.89
3.675	0.1225	0.1102	11.16
3.850	1.3141	1.1919	10.25
4.000	1.8605	1.4908	24.80
4.250	0.6753	0.6494	3.99
MAPE			10.32

**Fig. 11.** Simulation results of RMS output power and peak torque of the three-phase VREH as a function of the pitch at 10rpm (power) and 0.36rpm (torque).**Table 5**

Experimental results of power and torque, as well as power-to-torque ratio.

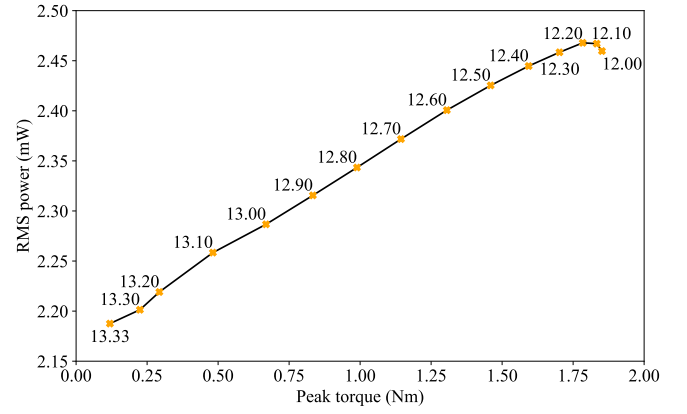
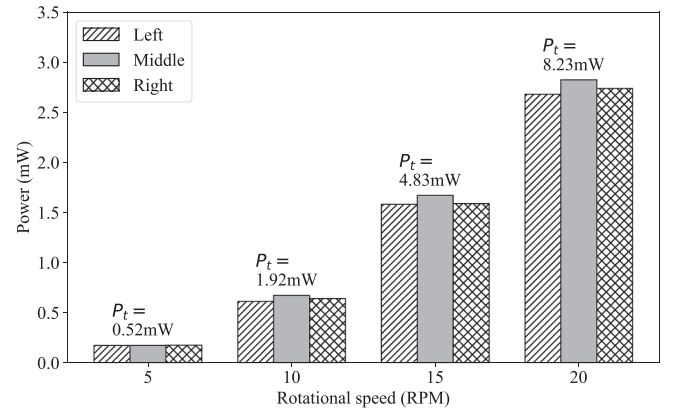
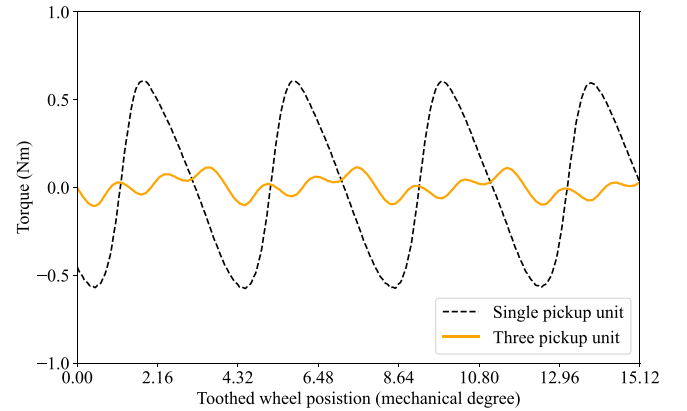
Pitch	Coil Power (uW)				Torque (Nm)	Ratio (uW/Nm)
	Left	Middle	Right	Total		
2.625	637.622	601.379	621.537	1860.537	0.364	5115.58
2.675	599.278	611.128	601.324	1811.731	0.110	16455.32
3.000	669.867	767.495	675.054	2112.416	1.622	1302.03
3.250	642.967	696.917	657.741	1997.625	0.610	3273.19
3.333	612.019	671.099	641.357	1924.476	0.115	16705.52
3.375	615.017	655.263	622.751	1893.032	0.341	5554.67
3.675	584.311	602.378	595.022	1781.711	0.110	16167.98
3.850	628.797	643.527	606.798	1879.121	1.192	1576.58
4.000	624.124	676.506	639.958	1940.588	1.491	1301.71
4.250	601.986	633.600	620.290	1855.876	0.649	2857.83

Note: Power is measured at 10rpm and torque at 0.36rpm

were in good agreement with the experiments, demonstrating a mean error of approximately 4% for output power and 10% for torque. Furthermore, the analytical, magnet-shifting method is able to accurately determine the optimal separation between the three pickup units.

The reduced cogging torque of the three-phase VREH can be explained based on phase shift between the individual pickup units, and thus their magnets. As a result, the cogging torque of one pickup unit can be compensated by that of the others when operating at an appropriate separation. In addition, part of the leakage flux of each pickup unit contributes to the flux change in the coil of their neighbors, which leads to the increased output power of the three-phase design.

Based on these results, it can be concluded that the proposed three-phase approach is a highly competitive, novel design for variable reluctance energy harvesters, outperforming single-pickup VREHs with

**Fig. 12.** Simulation results of maximum RMS output power as a function of acceptable peak torque.**Fig. 13.** Measured RMS output power of the optimized three-phase VREH, where P_t is the sum of the output power produced by the three individual coils (left, middle and right).**Fig. 14.** Experimental comparison of the torque ripple generated by VREHs with a single pickup unit and an optimized arrangement of three pickup units at 0.36rpm (2.16deg/s).

or without additional, torque-compensating magnets. The optimal separation between the three individual pickup units, however, was demonstrated to play a crucial role in the resulting performance, as both power and cogging torque are dependent on this relative position. Although the proposed analytical method cannot estimate absolute power and torque values, it can accurately estimate the optimal pickup unit arrangement, which leads to the highest power-to-torque ratio. A

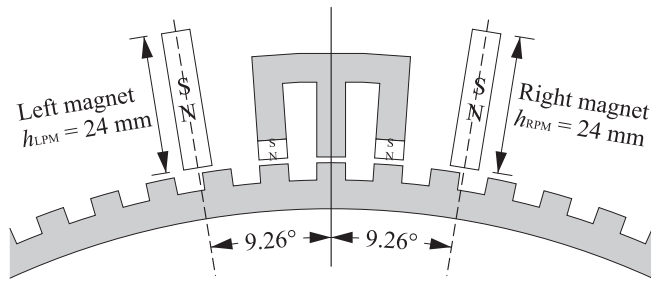


Fig. 15. The optimal structure of a single-phase pickup unit with two torque-compensating magnets. The two magnets with a height of $h_{LPM} = h_{RPM} = 24\text{mm}$ are placed in an arrangement of 9.26° ($= \text{pitch } 2.315$).

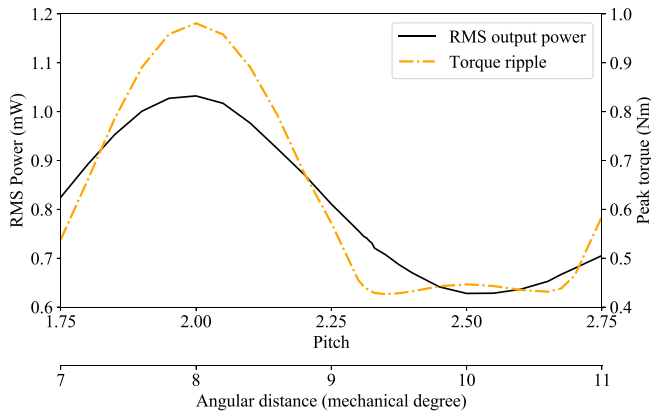


Fig. 16. Simulation results of RMS output power and peak torque of a single-phase VREH with two additional magnets ($h_{LPM} = h_{RPM} = 3.3\text{mm}$) as a function of the pitch at 10rpm (power) and the 0.36rpm (torque).

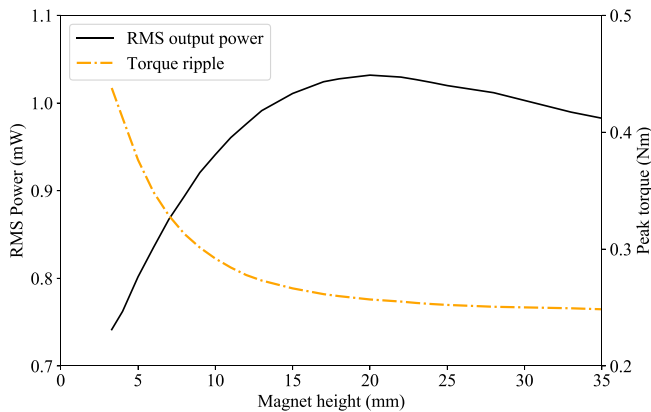


Fig. 17. Simulation results of RMS output power and peak torque of the single-phase VREH with two additional magnets at the optimal pitch of 2.315 as a function of the magnet height at 10rpm (power) and 0.36rpm (torque).

computational-intensive FEA is thus only needed to estimate the resulting torque and power values of the optimized configuration.

The performance gain of the three-phase VREH significantly improves the utility of wireless sensor systems operated by this type of energy harvester, resulting for example in higher sample rates, while reducing parasitic effects on the rotating host. As a result, this study contributes to improved energy harvester designs and their competitiveness to other technologies. Moreover, the proposed method and the knowledge generated by this study is not limited to the specific m-shaped structure investigated, but can be applied to guide the optimization of other variable reluctance transducers, both for energy harvesting and sensing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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